Correlation study of Lower hybrid amplitude saturation at Earth's magneto-pause

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Lower hybrid (LH) waves are studied for applications in thermonuclear fusion and space weather. The amplitudes and instabilities of these LH waves have been studied using analytical and numerical methods. In this paper these analytical methods that resulted from that research are tested using empirical data from the Earth's magneto-pause, using statistical analyse. Other plasma parameters that are not present in the analytical models, were also tested because of their potential relevance to the LH wave amplitude saturation.

Introduction

In this paper

Background Theory

Plasma environment

A plasma is distinguished from a neutral gas by its collective behavior, governed by long-range electromagnetic forces. To characterize the plasma environment at the magnetopause, we must define the fundamental temporal and spatial scales that dictate particle dynamics. The most fundamental frequency is the electron plasma frequency (ω_{pe}), which defines the characteristic timescale of collective electrostatic oscillations in response to a charge imbalance:

$$\omega_{pe} = \sqrt{\frac{n_e e^2}{m_e \epsilon_0}} \tag{1}$$

In contrast, the cyclotron frequency (ω_c) represents the frequency at which a charged particle gyrates around a magnetic field line due to the Lorentz force:

$$\omega_c = \frac{q|B|}{m} \tag{2}$$

The ratio $\frac{\omega_p}{\omega_c}$ is a critical parameter in wave analysis; at the Earth's magnetopause, this ratio is typically large ($\gg 1$), implying that the plasma is dense enough for electrostatic effects to be significant relative to magnetic constraints.

The Debye length (λ_D) is the scale over which a plasma shields out electric potentials. For phenomena occurring at scales larger than λ_D, the plasma maintains quasi-neutrality:

$$\lambda_D = \sqrt{\frac{\epsilon_0 k_B T}{nq^2}} \tag{3}$$

The spatial equivalent of gyration is the Larmor radius (or gyroradius), r_L , which represents the radius of a particle's circular motion around the magnetic field:

$$r_L = \frac{mv_\perp}{q|B|} = \frac{v_\perp}{\omega_c} \tag{4}$$

A useful ratio is $\frac{\omega_p}{\omega_c} \approx \frac{r_L}{\lambda_D}$, when this is much larger than unity, the plasma acts unmagnetized, and when the ratio is much smaller than unity, the plasma acts highly magnetized.

LH waves

Lower hybrid (LH) waves are quasi-electrostatic oscillations in a magnetized plasma that occur in a frequency range intermediate between the ion and electron cyclotron frequencies [17]. These waves propagate nearly perpendicular to the ambient magnetic field $\vec{B}_0$. The resonance occurs when the wave frequency ω satisfies $\omega_{ci} \ll \omega \ll \omega_{ce}$, a regime where ions are effectively unmagnetized while electrons are strongly tied to the magnetic field lines [16].

Physically, the LH resonance arises from charge separation. While ions respond to the wave's electric field $\delta\vec{E}$ via straight-line inertial motion, the electrons undergo an $\vec{E} \times \vec{B}$ drift perpendicular to both the wave propagation vector $\vec{k}$ and $\vec{B}_0$. In the cold plasma approximation, the lower hybrid resonance frequency ω_{LH} is given by:

$$\omega_{LH}^2 = \frac{\omega_{ci}^2 + \omega_{pi}^2}{1 + \frac{\omega_{pe}^2}{\omega_{ce}^2}} \tag{5}$$

where ω_{ci} and ω_{ce} are the ion and electron cyclotron frequencies, and ω_{pi} and ω_{pe} are the respective plasma frequencies. To analyze how this resonance depends on environmental parameters such as plasma density n and magnetic field magnitude B, the expression can be written as:

$$\omega_{LH}^2 = \frac{Z^2 e^2}{m_i} \left(\frac{\frac{B^2}{m_i} + \frac{n}{\epsilon_0}}{1 + \frac{nm_e}{\epsilon_0 B^2}} \right) \tag{6}$$

From the cold magnetized plasma dispersion relation, the ratio of the perpendicular to parallel wavevector components near resonance is [15]:

$$\frac{k_\perp}{k_\parallel} = \sqrt{\frac{\omega_{ce}\omega_{ci}}{\omega_{LH}^2 - \omega^2}} \tag{7}$$

As the wave frequency approaches the LH resonance frequency ($\omega \to \omega_{LH}$), the ratio $k_\perp/k_\parallel \to \infty$. This implies that the wavevector $\vec{k}$ becomes almost entirely perpendicular to the magnetic field $\vec{B}_0$, and the phase velocity directed along the field lines becomes very small.

LH wave amplitude prediction use cases

LH waves are used in fusion devices, and they also have a significant effect on the behavior of space weather.

Use cases in fusion: Lower hybrid wave instability for understanding transport during implosion and post-implosion phases of rapidly pulsed theta pinched experiments[6].

Use cases in predicting space weather: Lower hybrid drift instabilities (LHDI) plays a significant role in magnetic reconnection. Understanding LH wave saturation is critical for determining whether these waves can provide enough anomalous resistivity to facilitate magnetic reconnection in the collisionless plasma of the magnetopause.

General plasma wave instability theory

This section is for the readers intuition and basic knowledge about plasma wave instabilities. Wave instabilities are a product of available energy in the medium. For plasma waves we know that $\vec{E} \propto e^{i(\vec{k}\cdot\vec{r}-\omega t)}$, using $\phi = -\int \vec{E} \cdot d\vec{r}$ we get:

$$\phi \propto e^{i(\vec{k}\cdot\vec{r}-\omega t)} \tag{8}$$

This means that we can treat ϕ in Fourier space and replace ∇ with $-ik$ and get:

$$\nabla^2\phi = -k^2\phi \tag{9}$$

If the ratio $\frac{\nabla^2\phi}{\phi}$ is positive, k is imaginary, resulting in spatial amplitude growth. Usually wave instabilities in plasma physics are studied by linearization which allows one to derive dispersion relations. When these dispersion relations result in a positive imaginary angular frequency, the wave will grow temporally.

Lower hybrid drift instability

Lower hybrid drift instability (LHDI) is a cross-field current instability. Drift instabilities are the result of strong pressure gradients in a magnetized plasma, which create relative motion between ions and electrons due to diamagnetic drift. While the lower hybrid wave exists in a uniform plasma, a source of the wave is the Lower Hybrid Drift Instability (LHDI) which exists when spatial gradients in plasma density n, temperature T, or magnetic field magnitude B are present. To analyze this instability, the "slab approximation" is commonly employed.

The standard slab model is described in Cartesian coordinates. The magnetic field is typically oriented along the z-axis, $\vec{B} = B_0(x)\hat{z}$, while the equilibrium (between the pressure and magnetic pressure) gradients (such as the density gradient ∇n) are directed along the x-axis. This configuration drives a diamagnetic drift in the y-direction (the "cross-field" direction). This drift velocity is given by:

$$\vec{v}_D = -\frac{\nabla p \times \vec{B}}{nq|B|^2} \approx \frac{v_{th}^2}{2\omega_c L_n}\hat{y} \tag{10}$$

where v_{th} is the thermal velocity. Because the direction of the diamagnetic drift velocity is dependent on the sign of the charge (q), ions and electrons drift in opposite directions, generating a net diamagnetic current. This current drives a two stream drift instability between the magnetized electrons and the ions, which causes the LH perturbations to grow.

The cold plasma dispersion relation for the LHDI is [19][20]:

$$1 - \frac{\omega_{pi}^2}{\omega^2} + \frac{\omega_{pe}^2}{\omega_{ce}^2} - \frac{\omega_{pe}^2}{kL_n\omega_{ce}(\omega - kV_{e,y})} = 0 \tag{11}$$

In the cold-plasma limit, the instability is strongest when the lower hybrid frequency ω_{LH} matches the drift frequency $kV_{e,y}$. The maximum growth rate $\gamma = Im(\omega)$ can be approximated as:

$$\gamma_{max} \approx \sqrt{\frac{V_{e,y}\omega_{LH}}{L_n}} \tag{12}$$

LH wave instability saturation

It has been shown that an extended analytical quasi-linear model of LHDI saturation is consistent with full-kinetic simulations results [13][12]. There are two ways that the LHDI can saturate; current relaxation (Equation 13) and ion trapping (Equation 14), these occur at low and high density gradients respectively[13].

$$S_{max} = 2\frac{m_e}{m_i}\frac{(\epsilon_n\rho_i)^2}{(1+\omega_{pe}^2/\omega_{ce}^2)}n_iT_i \tag{13}$$

$$S_{max} = \frac{2}{45\sqrt{\pi}}\frac{(\epsilon_n\rho_i)^5}{(1+\omega_{pe}^2/\omega_{ce}^2)}n_iT_i \tag{14}$$

Where S_{max} is the saturated power density, $\epsilon_n = \frac{1}{L_n} = \frac{\nabla n}{n}$, ρ_i is the ion Larmor radius, and T_i is the ion temperature.

Current relaxation

LHDW create perturbations in the electric field (δE) and in the density (δn) which results in anomalous resistivity. The increased resistivity diffuses the strong magnetic field gradient in the current sheet and thus broadens(relaxes) it. Because of the equilibrium of solar wind pressure and magnetic pressure at the magneto-pause the broadening of the magnetic gradient also broadens the pressure gradient which is the source of instability for LHDW, thus creating a negative feedback loop, dampening the instability.

Wave particle trapping

For low magnetized plasma ($\omega_{ci} << \omega_{LH}$) ions have a low $\vec{E} \times \vec{B}$ drift compared to electrons, thus the trapping potential of ions for lower hybrid waves can be approximated as $v_T^2 = \frac{2e\phi}{m_i}$ where v_T is the maximum particle velocity in the frame of v_{ph} for it to be trapped in the wave and ϕ is the wave potential. The phase space of a hydrogen ion in an electrostatic wave in the frame of reference of the phase velocity is shown in figure 1. The red line shows the separatrix between trapped and untrapped particles. The trapped particles are oscillating in the potential well of the wave, while the untrapped particles are free to move through the wave. The trapping of particles in the wave potential leads to saturation of the wave amplitude, as they create an opposing potential.

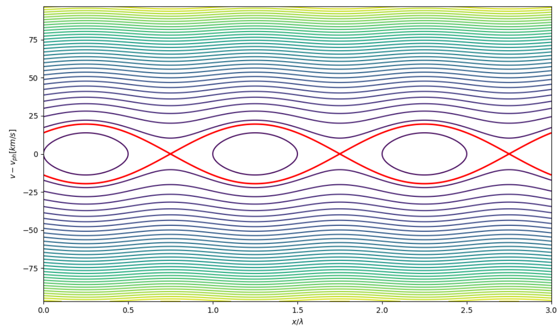

Figure 1: *Phase space of hydrogen ion in electrostatic wave in the frame of reference of phase velocity,* $A = 1[v/m]$.

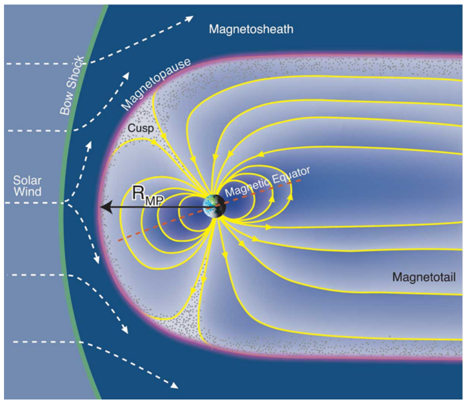

Figure 2: *Illustration of the magneto-pause boundary and the surrounding structures. Image adapted from the LASP MOP group website[2].*

Landau damping

The main difference between landau damping and damping due to wave particle trapping is that for the former small perturbations are assumed, the field is too small to significantly change the particle momentum, in other words; the former is linear and the latter is not. Landau wave particle interaction is a collisionless exchange of energy where particles slightly slower than the wave phase velocity are accelerated, and particles slightly faster are decelerated. Because a typical Maxwellian distribution has more slow particles than fast ones at the resonant velocity ($v \approx v_{ph} \rightarrow \frac{df(v)}{dv} < 0$), the wave experiences an overall loss of energy to the plasma.

Magneto-pause

The magneto-pause is the surface boundary between the magnetosheath and the magnetosphere. High energy ions an electrons in the solarwind hit the outer edge of the magnetosheath called the bowshock, from where the solar wind is compressed by earths magnetic field to form a plasma that is denser, slower and hotter then the upstream solar wind. The location of the magneto-pause is decided by the equilibrium of the dynamic solar wind pressure and the magnetic pressure of the earth's magnetic field. Therefore, the magneto-pause position R_{MP} oscillates and the MMS satellites record more than just two crossings per the elliptical orbit period.

$$R_m = \frac{vL}{\eta} \tag{15}$$

Where R_m is the magnetic Reynolds number, v is the plasma flow velocity, L is the characteristic flow scale and η is the magnetic diffusivity. For the plasma at the magnetosheath side of the magnetopause we know that $R_m >> 1$ thus the magnetic fields are "frozen" into the plasma flow due to a dominance of the induction term in the magnetic diffusion equation. When the plasma arrives at the magneto-pause L decreases to such an extent that magnetic diffusion will start dominating over the induction which allows for magnetic reconnection to occur. The magneto-pause is often not a smooth transition from the magnetosheath to the magnetosphere, but rather inhabited by structures such as current sheets and magnetic flux ropes [7].

Temperature

The temperature in a magnetized plasma is anisotropic due to the restrictions imposed on the movement of charged particles by the magnetic field.

Experimental Design

Parameter selection

The parameters that will be quantitatively compared with LH amplitudes to perform statistical tests are therefore:

Table 1: *Parameters for LH instability*

Parameter	Reason / relation to LH instability
Electron density n_e	Higher density lowers the LH frequency and can increase the available power for oscillations.
Electron density gradient ∇n_e	A high electron gradient is a source of available energy for plasma waves to grow.
Density gradient length scale L_n	
Drift current $J_{D\perp}$	The drift current perpendicular to the background field is a driver for LHDI.
Drift velocity $V_{e,\perp}$	Drift velocity in frame of reference of ions.
Electron Larmor radius $r_{L,e}$	
Ion Larmor radius $r_{L,i}$	
Plasma beta β	
Temperature ratio $\frac{T_i}{T_e}$	Ions are non-magnetized and cause Landau damping and are prone to wave particle trapping.
Ion temperature T_i	
Magnetic shear length scale L_s	Change of magnetic field direction damps LH waves because they only resonate perpendicular to that magnetic field.
Magnetic shear angle θ_S	
Potential perturbation $\delta\phi$	

We use n_e instead of n_i because the sampling for electrons is higher and the plasma should be quasi neutral.

Data acquisition

MMS

MMS consists of 4 satellites in a tetrahedral formation to differentiate between temporal- and spacial physical changes in the plasma. The NASA mission was launched in 2015. The formation follows an elliptical orbit around earth.

Density sensor

Electric field sensor

Magnetic field sensor

Table 2: *Instruments of MMS[8][10]. See references in Burch et al. (2015).*

Instrument	Quantity	Range	Rate (Burst mode)
Fast Plasma Investigation (FPI)	3D ion and electron flux distributions	10 eV–30 keV	e: 30 ms, ions: 150 ms
Fly's Eye Energetic Particle Spectrometer (FEEPS)	3D ion and electron flux distributions	25-500 keV (electrons) and 45-500 keV (ions)	10 s
Electric field Double Probes (EDP) and Axial Double Probes (ADP)	3D electric field	0.1-10 mV/m	8 Hz in slow survey, 32 Hz in fast survey, and 1024, 8192, or 65536 Hz in burst [11]
Electric Drift Instrument (EDI)			16 / 125 Hz
FluxGate Magnetometer (FGM)	3D magnetic field		128 Hz
Search Coil Magnetometer (SCM)	3D magnetic field		8192 / 16384 Hz

Measuring modes

fast brst etc

Magnetopause crossing detection

An algorithm using the density data to detect crossing was written in python. The algorithm uses the fact that the plasma density is significantly higher above the magneto-pause than below. It uses a threshold, which is chosen by manually finding some crossings and choosing an appropriate value

Data processing

The MMS database provides non scalar quantities in the GSE coordinate system which is defined as having $\hat{x}$ towards the Sun and its $\hat{z}$ normal to the plane of the Earth's orbit around the Sun, with positive z in the direction of Earth's magnetic north pole. These quantities were transformed into the field aligned coordinate (FAC) system where $\hat{z}$ is parallel to the local magnetic field vector. This was done by using the magnetic field data (in GSE) from FGM (Table 2).

As seen in Table 2 the DP electric field sensor in burst mode has the highest sampling rate. To compute quantities which are dependent on variables including the electric field, the other variables were up-sampled to that time resolution using interpolation.

The assumption was made that the only ion species is hydrogen. This is a common approximation for magnetopause crossings, as the magnetosheath plasma is dominated by solar wind protons. While heavier ions such as He^{2+} from the solar wind or O^+ from the ionosphere may be present (especially during significant geomagnetic activity) their number density is typically small compared to the proton population. The assumption was made to simplify calculations and because the ion composition data was not available for all events.

Divergence and curl

The gradient, curl and divergence of the fields are calculated with the assumption that the distances between the spacecraft are small compared to the distance of significant change; $d << \frac{X}{\nabla X}$ where d is the distance between spacecraft and X is an arbitrary quantity, such that the quantities can be assumed to be linear between the spacecraft.

These spatial derivatives were then calculated using the Reciprocal Vector Method [14], this method makes use of reciprocal vectors ($\vec{k}_\alpha$). These reciprocal vectors can be defined as follows. For a formation of four spacecraft at positions $\vec{r}_\alpha$ (where $\alpha = 1, 2, 3, 4$), we define the position of each probe relative to the center of mass as $\vec{x}_\alpha = \vec{r}_\alpha - \vec{r}_{CM}$. The reciprocal vectors satisfy the following.

$$\sum_{\alpha=1}^{4} \vec{k}_\alpha = 0, \sum_{\alpha=1}^{4} \vec{k}_\alpha \otimes \vec{x}_\alpha = \mathbf{I} \tag{16}$$

where $\mathbf{I}$ is the identity tensor. Each $\vec{k}_\alpha$ is a vector normal to the face of the tetrahedron opposite to spacecraft α , with a magnitude reciprocal to the height of the tetrahedron relative to that face. The spatial gradient operator ∇ is approximated using the reciprocal vectors as weights for the quantity measurements at each spacecraft location. For an arbitrary quantity X measured at the four spacecraft , the gradient tensor is calculated as:

$$\nabla X \approx \sum_{\alpha=1}^{4} \vec{k}_\alpha X_\alpha \tag{17}$$

Similarly for the curl:

$$\nabla \times \vec{X} \approx \sum_{\alpha=1}^{4} \vec{k}_\alpha \times \vec{X}_\alpha \tag{18}$$

The accuracy of this method is dependent on how well the MMS craft can maintain a tetrahedral formation, for the reciprocal vector will explode to infinity if the formation becomes planar to that face.

Parameter calculations

For the statistical analysis, plasma parameters must be accurately related to the LH wave amplitude. Since these parameters drive wave growth and determine saturation levels, they exhibit a dynamic, rather than instantaneous, correlation with the observed amplitude. Consequently, growth-related parameters and wave amplitudes rarely peak simultaneously. To account for this temporal offset, we sample data within a defined window around the LH amplitude peak: certain variables are averaged to represent the local environment, while for others, the maximum value within the interval is used.

The background drift is the total overall current in the plasma and can thus be calculated using Maxwell's equation taking displacement current to be zero, where we take the curl of the magnetic field $\nabla \times \vec{B} = \mu_0 \vec{J}$.

Using the assumption that the only ion species is hydrogen the current can be expressed as $J = e(n_i v_i - n_e v_e)$, setting the frame of reference of to the ion motion by taking $v_i = 0$ gives the expression for the cross field drift velocity as $V_{e,\perp} = -\frac{J_\perp}{n_e e}$. Like $J_\perp$, $V_{e,\perp}$ is a frame independent quantity because it is the difference of two velocities.

For the electron Larmor radius $r_{L,e}$ we need to know the velocity perpendicular to the magnetic field. The bulk velocity perpendicular to the magnetic field does not correspond to the scale of electron cyclotron oscillations, instead the thermal velocity is used. MMS's FPI and FEEPS sensors provide the electron distribution $f(\vec{r}, v, t)$ from which the second moment can be calculated;

$$P_{i,j}(\vec{r}, t) = m \int (v_i - V_i)(v_j - V_j) f(\vec{r}, v, t) d^3 v \tag{19}$$

Where m is the particle mass, v is the velocity, V is the bulk velocity (the first moment), P is the pressure, and i, j represent the orthogonal basis. To get $T_\perp$ we use $T = \frac{P}{n}$ where T is in energy units, by transforming the tensor to FAC one can get the perpendicular part $T_\perp = \frac{T_{xx+T_{yy}}}{2}$. The thermal velocity perpendicular to $\vec{B}$ now simply is $v_{th,\perp} = \sqrt{\frac{2T_\perp}{m}}$, and the larmor radius can now be expressed as $r_{L,e} = \frac{v_\perp m}{q|B|} = \frac{\sqrt{2T_\perp m}}{q|B|}$.

In a magnetized plasma, the pressure is anisotropic, meaning that β is anisotropic as well.

$$\beta_\perp = \frac{P_\perp 2\mu_0}{|B|^2}, \beta_\| = \frac{P_\| 2\mu_0}{|B|^2} \tag{20}$$

Here $P_\perp = \frac{Pxx+Pyy}{2}$ is the perpendicular pressure. Because the indication of pressure over magnetic pressure is of interest, the average of the parallel and perpendicular beta is used.

$$\beta = \frac{\beta_\| + 2\beta_\perp}{3} \tag{21}$$

The magnetic shear and its length scale are given by the following equations.

$$\epsilon_B = \frac{\nabla|\vec{B}|}{|\vec{B}|}, L_S = \frac{1}{\epsilon_B} \tag{22}$$

For the magnetic shear angle we need to know the magnetic field before and after the magneto-pause crossing.

$$cos(\theta_s) = \hat{B}_1 \cdot \hat{B}_2 = \frac{\vec{B}_1 \cdot \vec{B}_2}{|\vec{B}_1||\vec{B}_2|} \tag{23}$$

Instead of integrating the electric field the potential for the LH wave can be more easily computed using: $\delta\phi_B = \frac{|\vec{B}|}{en_e\mu_0}\delta B_\|$ [1]. To compute $\delta B_\|$ high time resolution data is needed, to bandpass filter it and get the parallel component to the background field. Therefore SCM (see table2) data was used for this parameter.

Case study

To show the analysis of the environment data for each magneto-pause crossing event, this paragraph goes into depth about just one specific crossing event by investigating the resulting plasma parameters. The crossing event chosen for the case study is at the time interval [2015-11-21 1:45:59.206, 2015-11-21 1:46:59.206], and seems to be a crossing from the magnetosphere to the magnetosheath. In this section the LH amplitude peak references to the peak of the bandpass filtered $E_\perp$ amplitude.

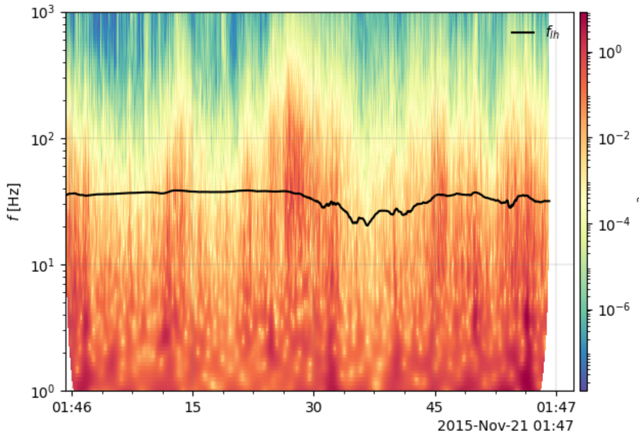

Figure 3: *$E\perp$ spectrogram with theoretical LH frequency from plasma environment characteristics.*

Figure 3 shows the spectrogram of $E_\perp$ with the theoretical LH frequency calculated from the plasma parameter overlaid.

The LH frequency is not exactly at the theoretical frequency because of multiple reasons: The spectrum can be shifted due to Doppler shift, because the magnetic field lines are not stationary and neither are the mms spacecraft. When they travel parallel to $\vec{k}$ there will be a dopplershift.

LH drift instability, this instability is non-linear and when the waves grow they decay into other frequencies.

The theoretical LH frequency used in Figure 3 is derived using the cold plasma assumption.

And finaly, the assumption was made that only hydrogen ions are present, if other ions are present the actual LH frequency will be lower.

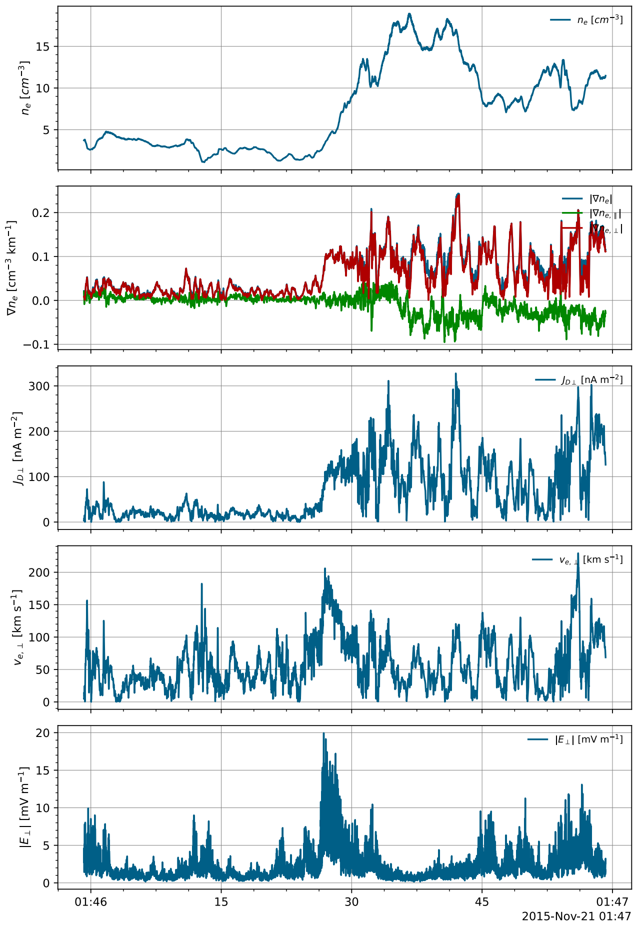

Figure 4: *Locally measured quantities averaged over the 4 MMS spacecraft.*

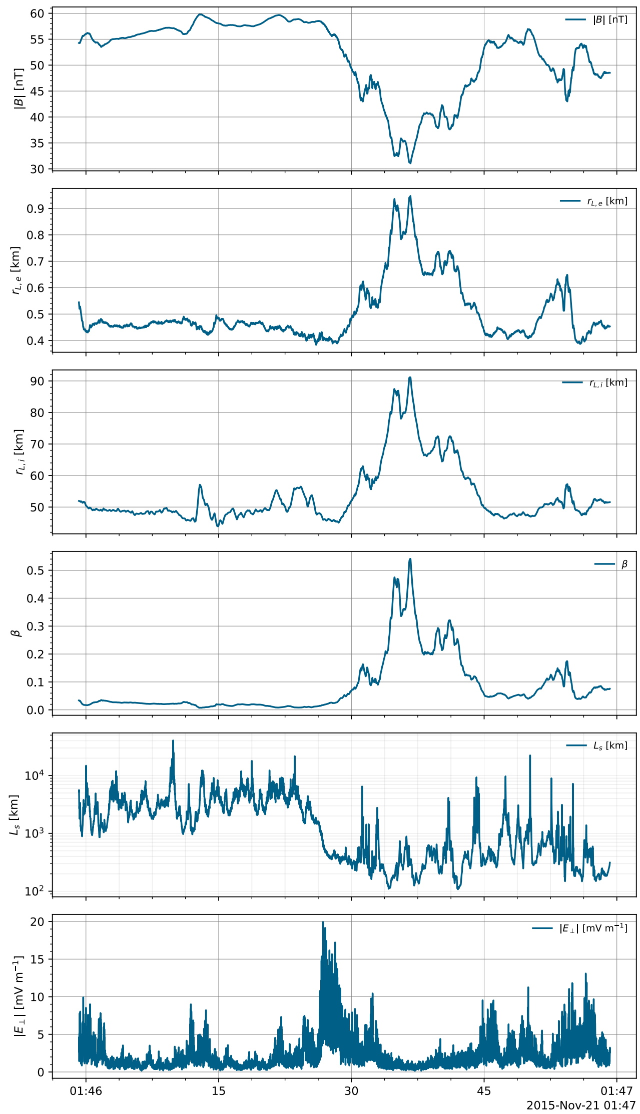

Figure 5: *Locally measured quantities averaged over the 4 MMS spacecraft.*

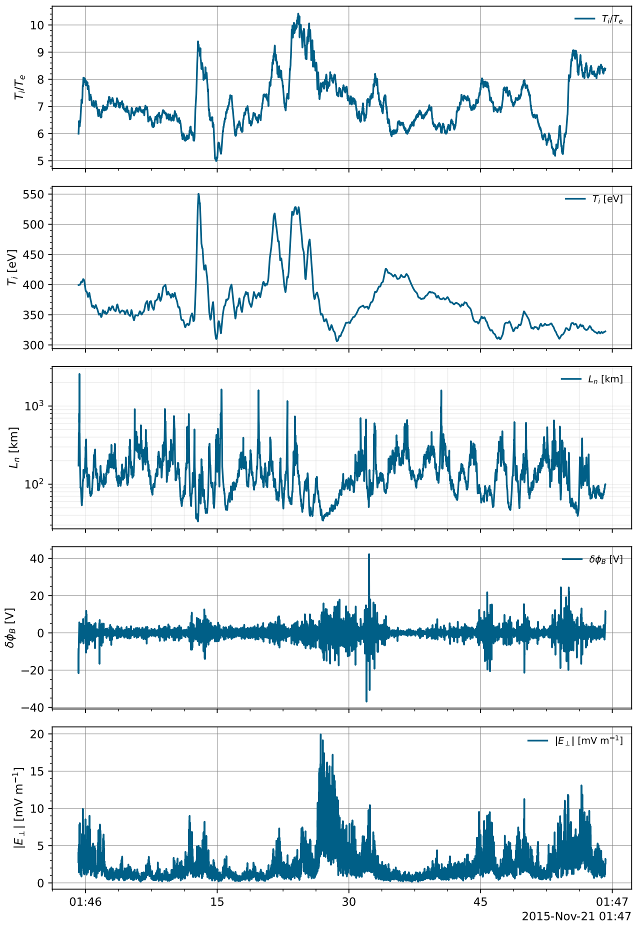

Figure 6: *Locally measured quantities averaged over the 4 MMS spacecraft.*

Figure 4 shows the electron density, as one would expect; there is a significant increase in density at the determined magneto-pause crossing event. The representative value for the statistical analysis is taken as the average value in an interval around the LH amplitude peak.

Figure **??** shows the magnetic field magnitude, as expected; the magnitude decreases when the spacecraft leave the magnetosphere.

Figure **??** shows the current density calculated using the curl of the magnetic field. There is a large increase in cross field current density at the magneto-pause crossing, as one expects when going from a high to low magnetic pressure region. The representative value for the statistical analysis of the current density is taken as the maximum value in an interval around the LH amplitude peak.

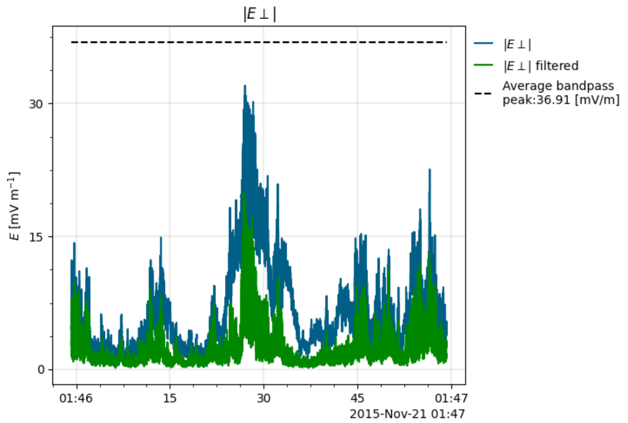

Figure 7: *Perpendicular E field bandpass filtered to [5,200] Hz. The amplitude of the LH waves is taken as the local maximum of this signal. The plotted values are the averaged values over the 4 spacecraft.*

Figure 7 shows the bandpass filtered perpendicular E field. The amplitude taken for this crossing event is the average of the peaks from the 4 MMS spacecraft.

Figure **??** shows the gradient of the electron density. As expected the gradient increases to a new equilibrium value around the crossing event, which indicates that the crossing algorithm works as intended. The maximum value of the gradient in an interval around the LH amplitude peak is taken as the representative value for this crossing event.

Figure **??** shows the density gradient length scale. It shows a lower length scale of significant change of density on the magnetosphere side, which drives LHDI. The representative value for the statistical analysis is taken as the minimum over an interval around the LH amplitude peak.

Figure **??** shows the cross field drift velocity in the reference frame of the ions. A broad peak can be seen around the magneto-pause crossing this is likely due to the high gradients ∇B and ∇n in the magneto-pause boundary, because they drive drift currents. The maximum value in an interval around the LH amplitude peak is taken as the representative value.

Note that the ion Larmor radius is two orders of magnitude larger than the electron Larmor radius, this relates to the the different behavior of electrons and ions mentioned in the introduction. This is why electrons behave as magnetized particles and ions do not.

The time interval for averaging the parameters was chosen to be 0.5 second before and after the LH amplitude peak. This interval was chosen after manually looking at a handful of crossing events and deciding that this is long enough to capture the local environment, but short enough to not include other events that might influence the parameters. For finding the maximum values of the parameters, it was decided in the same manner to take an interval of 15 seconds before and after the LH amplitude peak. For the saturation values of the parameters, it was decided to take an interval of $\frac{1}{10}$ seconds before the LH amplitude peak. Because the saturation values before the measurement actually influence it and the saturation time is inverse propotional to the LH frequancy which lowest value was manually determined to be around 10 Hz.

Statistical analysis

To find correlations between the parameters and the LH amplitudes, statistical tests are performed. And to test the analytical models, fits are performed and analyzed.

Normalization

To compare results from different crossing events it is helpful to normalize certain parameters, making them dimensionless. For example the lengthscales were normalized using the LH scale, this gives a dimensionless quantity that can be compared between different crossings.

$$\hat{L}_n = \frac{L_n}{\sqrt{r_{L,e} r_{L,i}}}, \hat{L}_s = \frac{L_s}{\sqrt{r_{L,e} r_{L,i}}} \tag{24}$$

$$\epsilon_n \rho = \epsilon_n \sqrt{r_{L,e} r_{L,i}}, \epsilon_s \rho = \epsilon_s \sqrt{r_{L,e} r_{L,i}} \tag{25}$$

Furthermore the ratio of $\frac{L_n}{L_s}$ is investigated, because it provides a dimensionless ratio of instability driving and damping quantities. The wave potential is relevant in this study due to its relation between wave trapping and landau damping. Therefore it is normalized using the ion and electron thermal energy. So that it is related to the particles likely-hood of escaping the wave's potential trap.

$$\hat{\Phi}_e = \frac{e\delta\phi}{T_e}, \hat{\Phi}_i = \frac{e\delta\phi}{T_i} \tag{26}$$

The drift velocity is normalized using the ion thermal velocity: $\frac{V_{e,\perp}}{v_{i,th,\perp}}$, because the analytical derivations growth rates and saturation levels are dependent on this ratio [6]. For the temperature the ratio $\frac{T_i}{T_e}$ is used, because theory suggests T_i and T_e have a positive and negative correlation with the LH amplitude respectively.

$$\frac{V_{e,\perp}}{v_A}, v_A = \frac{B}{\sqrt{\mu_0 n_i m_i}} \tag{27}$$

$$\frac{V_{e,\perp}}{v_{th,i}}, v_{th,i} = \sqrt{\frac{2T_i}{m_i}} = r_{L,i}\omega_{ci} \tag{28}$$

For $\frac{T_i}{T_e}$ much greater than 1, so electron diamagnetic drifts are much smaller than ion diamagnetic drifts, $V_\perp / v_{th,i}$ should be similar to $\epsilon_n \rho_i$. Therefore, one might expect that it has similar importance to $\epsilon_n \rho_i$.

$$\frac{\omega_{pe}}{\omega_{ce}} \tag{29}$$

Statistical tests

Spearman's correlation test is used to find correlations between the parameters and the LH amplitudes. Because the correlations are likely not linear and it tests for general monotonicity. Partial correlation is needed as well, because the LH frequencies are dependent on parameters that are also dependent on each other.

Random Forest feature importance

For a better insight in which features are most important to predict the LH amplitude, a random forest model was employed. A random forest model is a machine learning algorithm

Comparison with theoretical models

The electric field amplitude of the LH waves is related to the saturated power density using $W_E = \frac{1}{2}\epsilon_0 |\vec{E}|^2$, where W_E is the electric field energy density which is equal to S_{max} (equation 13 and 14) at saturation.

Analysis results

The magneto-pause crossing detection algorithm detected 4260 crossing with available high time resolution (burst mode) data, after computing forementioned plasma parameters the data was sanitized to remove invalid entries (Nan and inf) that resulted from erros such as devision by near-zero gradients.

Table 3: *Parameters Spearman correlation coefficients and p-values with respect to the LH amplitude.*

Parameter	Correlation coefficient	P value
n_e	-0.014	0.439
∇n_e	0.377	< 0.001
$J_{D\perp}$	0.474	< 0.001
$V_{e,\perp}$	0.528	< 0.001
$\frac{v_e}{v_i}$	0.377	< 0.001
β	-0.064	< 0.001
$\frac{T_i}{T_e}$	0.086	< 0.001
T_i	0.332	< 0.001
L_s	-0.224	< 0.001
L_n	-0.411	< 0.001
$\epsilon_n \rho_i$	0.354	< 0.001
$\frac{\omega_{pe}^2}{\omega_{ce}^2}$	-0.206	< 0.001
B	0.288	< 0.001
θ_S	0.068	< 0.001
$\frac{v_e}{v_{th,i}}$	0.349	< 0.001
$v_{th,i}$	0.338	< 0.001

Table 4: *Parameters Spearman correlation coefficients and p-values with respect to the perturbed potential.*

Parameter	Correlation coefficient	P value
n_e	-0.279	< 0.001
∇n_e	0.053	0.004
$J_{D\perp}$	0.303	< 0.001
$V_{e,\perp}$	0.605	< 0.001
$\frac{v_e}{v_i}$	0.274	< 0.001
β	-0.185	< 0.001
$\frac{T_i}{T_e}$	0.207	< 0.001
T_i	0.581	< 0.001
L_s	-0.111	< 0.001
L_n	-0.264	< 0.001
$\epsilon_n \rho_i$	0.264	< 0.001
$\frac{\omega_{pe}^2}{\omega_{ce}^2}$	-0.382	< 0.001
B	0.290	< 0.001
θ_S	0.154	< 0.001
$\frac{v_e}{v_{th,i}}$	0.258	< 0.001
$v_{th,i}$	0.580	< 0.001

Correlation does not mean causation or that the quantiy is a driver for LH instability. n_e and L_n show a positive correlation for example because the LH waves need particles to carry current and need a threshold density to be excited.

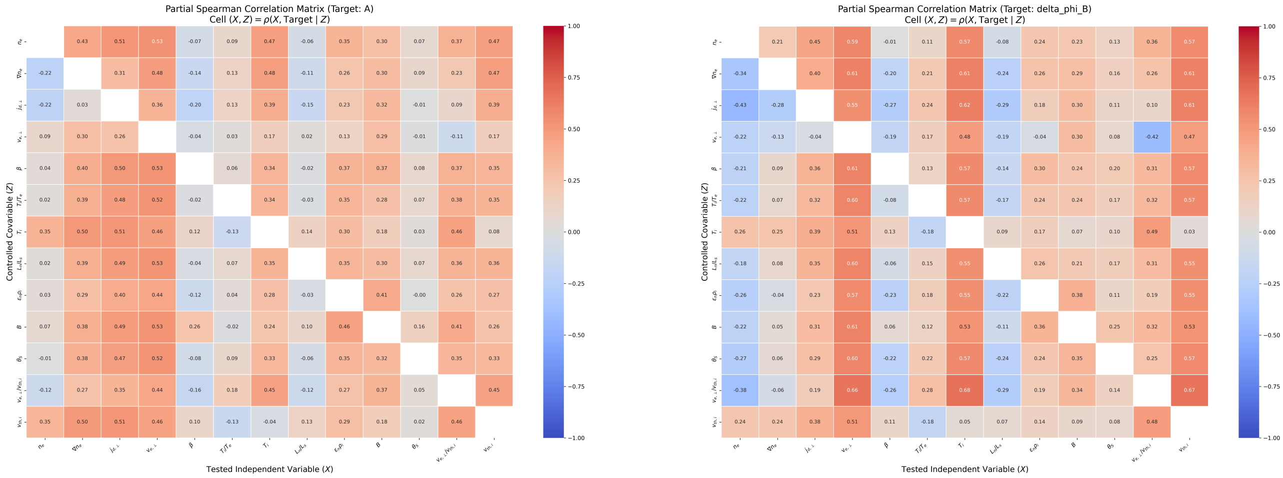

Figure 8: *captioon.*

Comparison with theoretical models

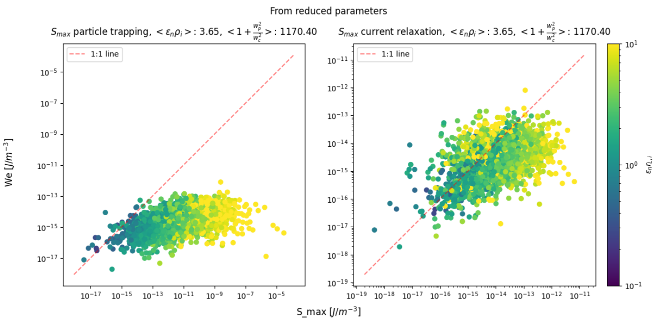

Figure 9: *Wave energy density saturation from LH quasi-linear model over the measured LH energy density calculated from reduced parameters.*

Both models should be used for a certain $\epsilon_n\rho_i$, in this project $\epsilon_n\rho_i > 1.25$ is used as the working definition for a strong gradient [12].

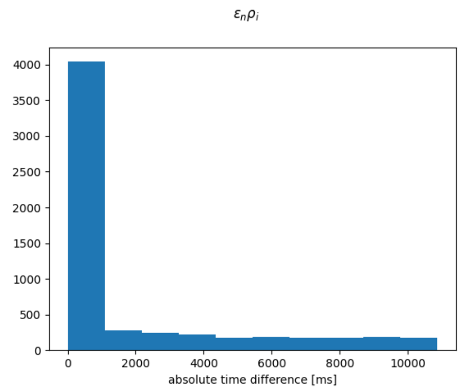

Figure 10

Figure 10 shows a flat distribution after a certain time, this could mean that these maxima of epsilon are unrelated to the maximum LH amplitude. We test this by filtering out all samples where this time diference is greater then 2000 ms.

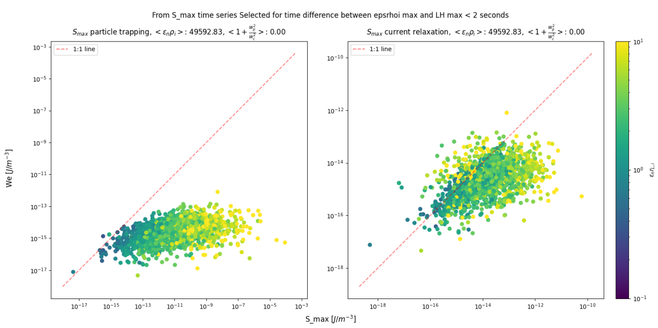

Figure 11

Conclusion

Discussion

The assumption that the only ion species is hydrogen might result in computational errors for significant geomagetic events.

The magneto-pause crossing detection algorithm used for this study only selects crossing events with a certain change in density, this means the dataset is biased towards crossings with a somewhat significant density gradient. Thus the statistical test results may find a lower feature importance for the density gradient.

Low correlation for the density gradient could be due to the fact that larger waves lower the gradient due to anomulous transport. The LH amplitude correlation results for the density gradient could also be due to the fact that when LH wave amplitudes are observations are done the density gradient is already lowered due to anomalous transport. In other words, the gradient responsible for the high wave amplitude may no longer be present when measurements occur.

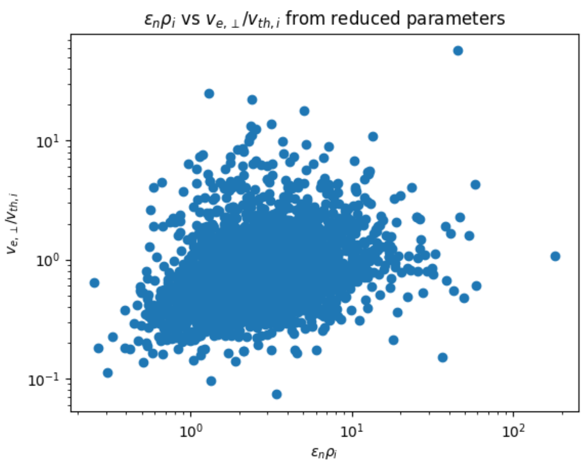

Figure 12

$\frac{v_{\perp,e}}{v_{th,i}} = \epsilon_n \rho_i$ when assuming that the only source of drift is the density gradient (diamagnetic drift). Why does figure **??** show a low importance for the normalized cross field drift even though $\epsilon_n \rho_i$ and $\frac{v_{e,\perp}}{v_{th,i}}$ should be similar, figure 12 shows a correlation but high variance ....

List of abbreviations

LH	Lower Hybrid
LHDW	Lower Hybrid Drift Waves
LHDI	Lower Hybrid Drift Instability
MMS	
GSE	Geocentric Solar Ecliptic coordinate system
FAC	Field Alligned Coordinate system

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Appendix A